

1. What is the correct extension of a C source file?

- A. .cpp
- B. .java
- C. .c
- D. .exe

Answer: C

2. Which data structure follows the FIFO principle?

- A. Stack
- B. Queue
- C. Tree
- D. Graph

Answer: B

3. What is the output of 5 / 2 in C (integer division)?

- A. 2.5
- B. 2
- C. 3
- D. Error

Answer: B

4. Which loop executes at least once?

- A. for
- B. while
- C. do-while
- D. None

Answer: C

5. What does HTML stand for?

- A. Hyper Text Markup Language
- B. High Text Machine Language
- C. Hyper Transfer Markup Language
- D. Hyper Tool Markup Language

Answer: A

6. Which keyword is used to define a function in Python?

- A. function
- B. define
- C. func
- D. def

Answer: D

7. Which data type stores True or False values?

- A. int
- B. char
- C. bool
- D. float

Answer: C

8. Which operator is used for comparison in Python?

- A. =
- B. ==
- C. :=
- D. =>

Answer: B

9. What is the time complexity of binary search?

- A. $O(n)$
- B. $O(n^2)$
- C. $O(\log n)$
- D. $O(1)$

Answer: C

10. Which data structure uses LIFO?

- A. Queue
- B. Stack
- C. Linked List
- D. Array

Answer: B

11. Which symbol is used for comments in Python?

- A. //
- B. /* */
- C. #
- D. --

Answer: C

12. Which language is primarily used for Android development?

- A. Java
- B. COBOL
- C. Pascal
- D. Perl

Answer: A

13. Which sorting algorithm repeatedly swaps adjacent elements?

- A. Merge Sort
- B. Bubble Sort
- C. Quick Sort
- D. Heap Sort

Answer: B

14. What is the output of len("Hello") in Python?

- A. 4
- B. 5
- C. 6
- D. Error

Answer: B

15. Which of the following is not a programming language?

- A. Python
- B. Java
- C. Linux
- D. C++

Answer: C

16. What is the value of `2 ** 3` in Python?

- A. 6
- B. 8
- C. 9
- D. 5

Answer: B

17. Which data structure stores elements in key-value pairs?

- A. Array
- B. Dictionary
- C. Stack
- D. Queue

Answer: B

18. Which keyword is used to exit a loop?

- A. stop
- B. continue
- C. break
- D. return

Answer: C

19. Which of the following is a relational database?

- A. MongoDB
- B. Redis

- C. MySQL
- D. Cassandra

Answer: C

20. What does SQL stand for?

- A. Structured Query Language
- B. Standard Query Logic
- C. Sequential Query Language
- D. System Query Language

Answer: A

21. Which symbol is used to access members of an object in Java?

- A. #
- B. @
- C. .
- D. ->

Answer: C

22. Which algorithm is used to find the shortest path in a graph?

- A. Bubble Sort
- B. Dijkstra's Algorithm
- C. Binary Search
- D. DFS

Answer: B

23. What is recursion?

- A. A loop structure
- B. A function calling itself
- C. A sorting method
- D. A database query

Answer: B

24. Which keyword creates an object in Java?

- A. class
- B. create
- C. object
- D. new

Answer: D

25. Which of these is a valid Python list?

- A. {1,2,3}
- B. (1,2,3)
- C. [1,2,3]
- D. <1,2,3>

Answer: C

26. What is the output of $10 \% 3$?

- A. 3
- B. 1
- C. 0
- D. 10

Answer: B

27. Which searching technique requires sorted data?

- A. Linear Search
- B. Binary Search
- C. DFS
- D. BFS

Answer: B

28. Which OOP concept allows one class to acquire properties of another?

- A. Encapsulation
- B. Polymorphism
- C. Inheritance
- D. Abstraction

Answer: C

29. What is the default return type of a function in Python without a return statement?

- A. 0
- B. False
- C. Null
- D. None

Answer: D

30. Which of the following is used to store data permanently?

- A. RAM
- B. Cache
- C. Hard Disk / SSD
- D. Register

Answer: C